04_profiles naming the segments

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1. Goal

Take the k = 3 cluster labels from and turn them into named, interpretable customer personas. Output: `../data/customer_features_labeled.csv`.

2. Load + validate

```
assignments <- read_csv("../data/processed/customer_clusters.csv",
                        show_col_types = FALSE)
customer    <- read_csv("../data/processed/customer_features.csv",
                        show_col_types = FALSE)
tx          <- read_csv("../data/interim/online_retail_clean.csv",
                        show_col_types = FALSE)
cancelled   <- read_csv("../data/interim/online_retail_cancelled.csv",
                        show_col_types = FALSE)

tx$InvoiceDate        <- as_datetime(tx$InvoiceDate)
cancelled$InvoiceDate <- as_datetime(cancelled$InvoiceDate)
tx <- tx |>
  mutate(Description = iconv(Description, from = "", to = "UTF-8", sub = ""))

stopifnot(
  "CustomerID" %in% names(assignments),
  "CustomerID" %in% names(customer),
  "CustomerID" %in% names(tx),
  !anyDuplicated(customer$CustomerID),
  setequal(assignments$CustomerID, customer$CustomerID)
)

cust <- customer |>
  inner_join(assignments |> select(CustomerID, cluster),
             by = "CustomerID") |>
  mutate(cluster = factor(cluster))

list(customers_clustered = nrow(cust),
     k                   = n_distinct(cust$cluster),
     cluster_sizes       = table(cust$cluster))
```

```
## $customers_clustered
## [1] 4334
##
## $k
## [1] 3
##
## $cluster_sizes
##
##    A    B    C
## 1396 1804 1134
```

2a. Re-label clusters in stable A → B → C order

K-means cluster IDs depend on the random seed. Re-label by descending median `log_NetMonetary` so cluster `A` is always the highest-value group, `C` the lowest, regardless of the seed.

```
order_by_value <- cust |>
  group_by(cluster) |>
  summarise(med = median(log_NetMonetary), .groups = "drop") |>
  arrange(desc(med)) |>
  mutate(label = LETTERS[seq_len(n())]) |>
  select(cluster, label)

cust <- cust |>
  left_join(order_by_value, by = "cluster") |>
  mutate(cluster = factor(label, levels = LETTERS[seq_along(unique(label))])) |>
  select(-label)

tx <- tx |>
  inner_join(cust |> select(CustomerID, cluster), by = "CustomerID")

table(cust$cluster)
```

```
##
##    A    B    C
## 1396 1804 1134
```

3. Headline summary share of customers vs share of revenue

```
total_rev_net <- sum(cust$NetMonetary)

headline <- cust |>
  group_by(cluster) |>
  summarise(
    customers         = n(),
    revenue_net       = sum(NetMonetary),
    revenue_gross     = sum(GrossMonetary),
    median_net_spend = median(NetMonetary),
    median_orders     = median(Frequency),
    median_recency_d = median(Recency),
```

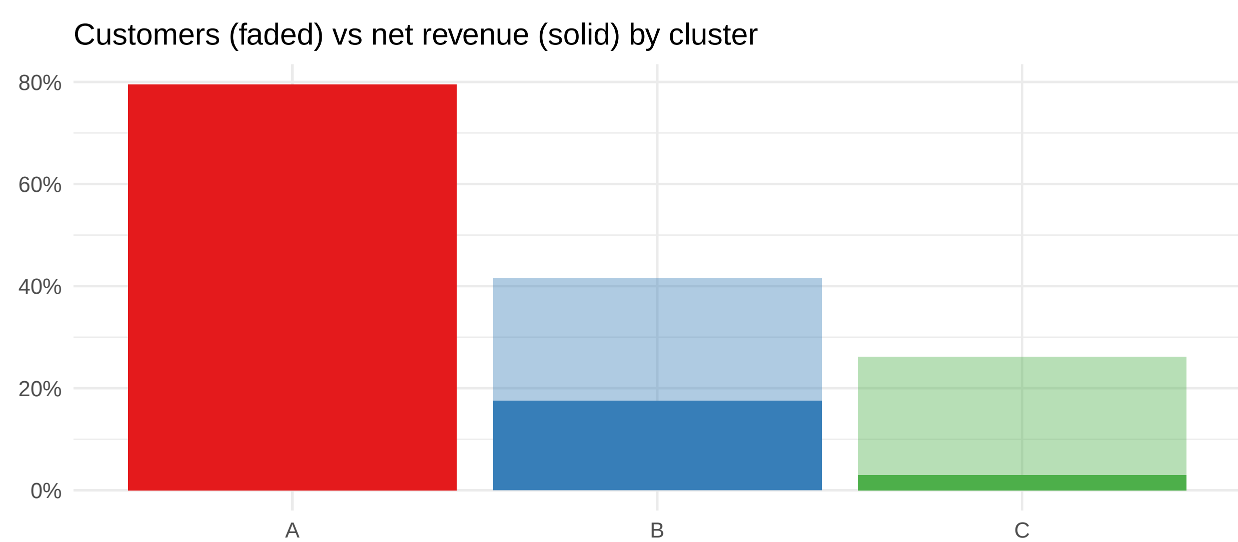
```
    .groups = "drop"
  ) |>
  mutate(pct_customers = customers   / sum(customers),
         pct_revenue   = revenue_net / total_rev_net) |>
  arrange(cluster)

headline |>
  mutate(across(c(revenue_net, revenue_gross, median_net_spend),
                \(x) dollar(x, prefix = pound)),
         across(c(pct_customers, pct_revenue),
                \(x) percent(x, accuracy = 0.1))) |>
  kable(caption = "Cluster headlines   share of customers vs revenue (net)")
```

Table 1: Cluster headlines share of customers vs revenue (net)

cluster	customers	revenue_net	revenue_gross	median_net_spend	median_orders	median_recency	pct_customers	pct_revenue
A	1396	£6,574,443	£6,734,530	£2,264.95	6	16.0	32.2%	79.5%
B	1804	£1,447,805	£1,741,381	£600.46	2	70.5	41.6%	17.5%
C	1134	£247,152	£261,317	£178.61	1	156.0	26.2%	3.0%

The most important number above is **share of revenue vs share of customers**. If cluster A holds ~28% of customers but ~75% of net revenue, that's the segment to protect; the inverse pattern in cluster C identifies the segment to either nurture or accept as low-margin.

```
ggplot(headline, aes(cluster, fill = cluster)) +
  geom_col(aes(y = pct_customers), alpha = 0.4, position = "identity") +
  geom_col(aes(y = pct_revenue),   alpha = 1.0, position = "identity") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Customers (faded) vs net revenue (solid) by cluster",
       y = NULL, x = NULL) +
  theme(legend.position = "none")
```



4. Feature profile median AND mean

Median is robust to outliers; mean is what aggregates revenue forecasts. We report both because the distributions are heavy-tailed.

```r
profile <- cust |>
  group_by(cluster) |>
  summarise(
    n                    = n(),
    Recency_med          = median(Recency),
    Recency_mean         = mean(Recency),
    Frequency_med        = median(Frequency),
    Frequency_mean       = mean(Frequency),
    NetMonetary_med      = median(NetMonetary),
    NetMonetary_mean     = mean(NetMonetary),
    AvgOrderValue_med    = median(AvgOrderValue),
    AvgOrderValue_mean   = mean(AvgOrderValue),
    UniqueProducts_med   = median(UniqueProducts),
    UniqueProducts_mean  = mean(UniqueProducts),
    AvgUnitPrice_med     = median(AvgUnitPrice_winsor),
    TenureDays_med       = median(TenureDays),
    SinglePurchase_share = mean(SinglePurchase),
    ReturnRate_med       = median(ReturnRate),
    ReturnRate_mean      = mean(ReturnRate),
    ReturnValueRate_med  = median(ReturnValueRate),
    ReturnValueRate_mean = mean(ReturnValueRate),
    .groups = "drop"
  )

profile |>
  mutate(across(c(NetMonetary_med, NetMonetary_mean,
                  AvgOrderValue_med, AvgOrderValue_mean,
                  AvgUnitPrice_med),
                \(x) dollar(x, prefix = pound, accuracy = 0.01)),
         across(c(SinglePurchase_share,
                  ReturnRate_med, ReturnRate_mean,
                  ReturnValueRate_med, ReturnValueRate_mean),
                \(x) percent(x, accuracy = 0.1))) |>
  kable(caption = "Median and mean profile per cluster (original scales)")
```

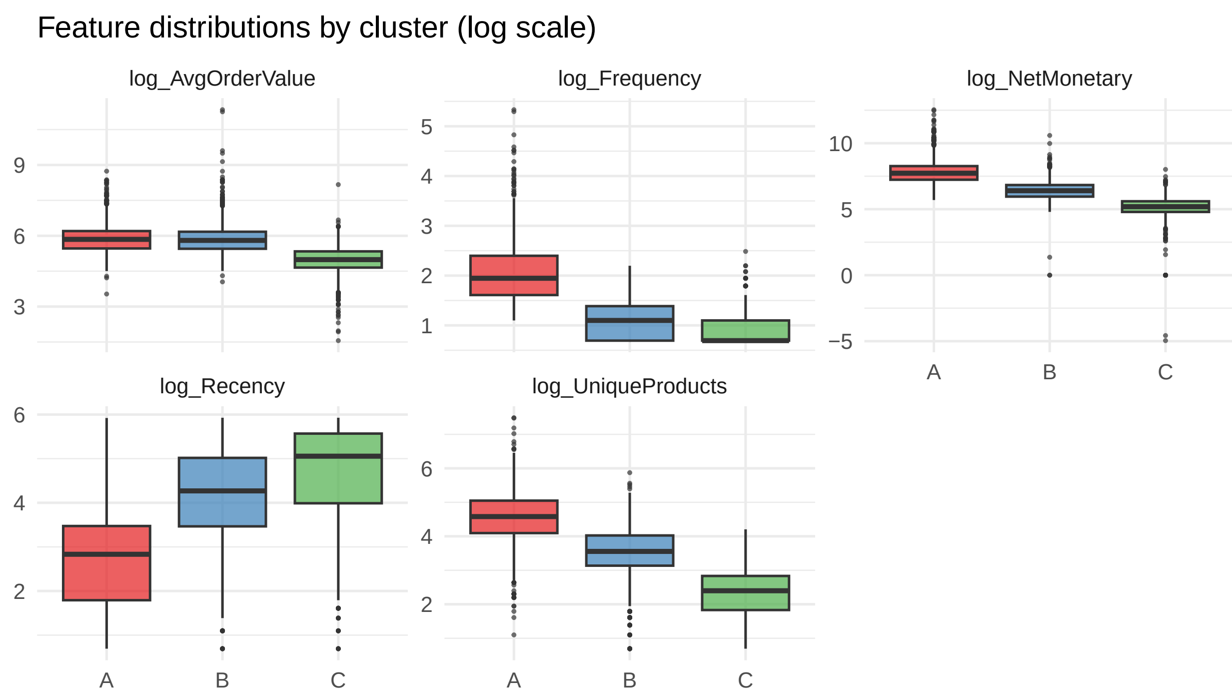
Table 2: Median and mean profile per cluster (original scales)

cluster	n	Recency_med	Recency_mean	Frequency_med	Frequency_mean	NetMonetary_med	NetMonetary_mean	AvgOrderValue_med	AvgOrderValue_mean	UniqueProducts_med	UniqueProducts_mean	AvgUnitPrice_med	TenureDays_med	SinglePurchase_share	ReturnRate_med	ReturnRate_mean	ReturnValueRate_med	ReturnValueRate_mean
A	1396	16.0	24.58736	9	9.340974	£2,264.95	£4,709.34	£345.49	£450.98	86.5	125.1346	£17.77	293	0.0%	12.5%	13.7%	0.5%	2.0%
B	1804	70.5	102.0652	41	2.068188	£800.46	£802.55	£330.98	£540.96	64.0	42.99390	£11.67	41	38.1%	0.0%	9.6%	0.0%	1.5%
C	1134	156.0	161.6621	26	1.438272	£278.61	£217.95	£145.91	£172.17	10.0	12.32099	£92.18	0	72.1%	0.0%	6.5%	0.0%	3.2%

5. Distributions by cluster

A median can mislead when distributions are skewed. Boxplots on the log scale show the spread within each cluster.

```r
cust |>
  select(cluster, log_Recency, log_Frequency, log_NetMonetary,
         log_AvgOrderValue, log_UniqueProducts) |>
  pivot_longer(-cluster, names_to = "feature", values_to = "value") |>
  ggplot(aes(cluster, value, fill = cluster)) +
  geom_boxplot(alpha = 0.7, outlier.size = 0.4) +
  facet_wrap(~ feature, scales = "free_y") +
  scale_fill_brewer(palette = "Set1") +
  labs(title = "Feature distributions by cluster (log scale)",
       x = NULL, y = NULL) +
  theme(legend.position = "none")
```

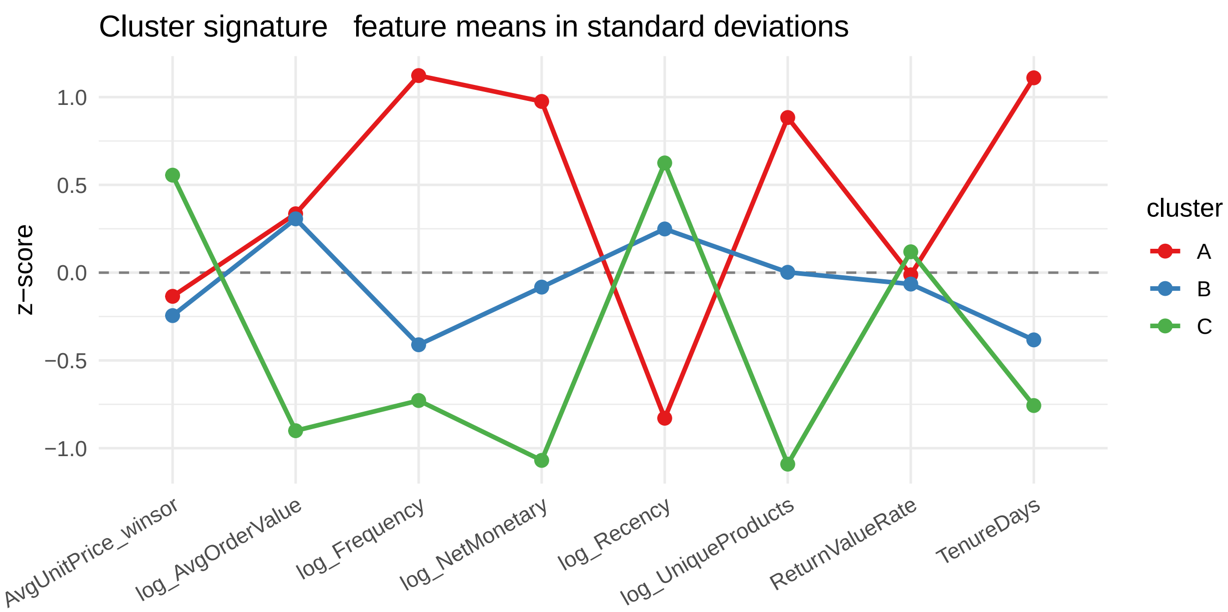


6. Parallel-coordinate signature

A single picture summarising "what makes each cluster different from the average customer." Each line is a cluster centroid in z-scored feature space positive = above population mean, negative = below.

```r
z_profile <- cust |>
  select(cluster, log_Recency, log_Frequency, log_NetMonetary,
         log_AvgOrderValue, log_UniqueProducts,
         AvgUnitPrice_winsor, TenureDays, ReturnValueRate) |>
  mutate(across(-cluster, \(x) as.numeric(scale(x)))) |>
  group_by(cluster) |>
  summarise(across(everything(), mean), .groups = "drop") |>
  pivot_longer(-cluster, names_to = "feature", values_to = "z")

ggplot(z_profile, aes(feature, z, group = cluster, colour = cluster)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
```

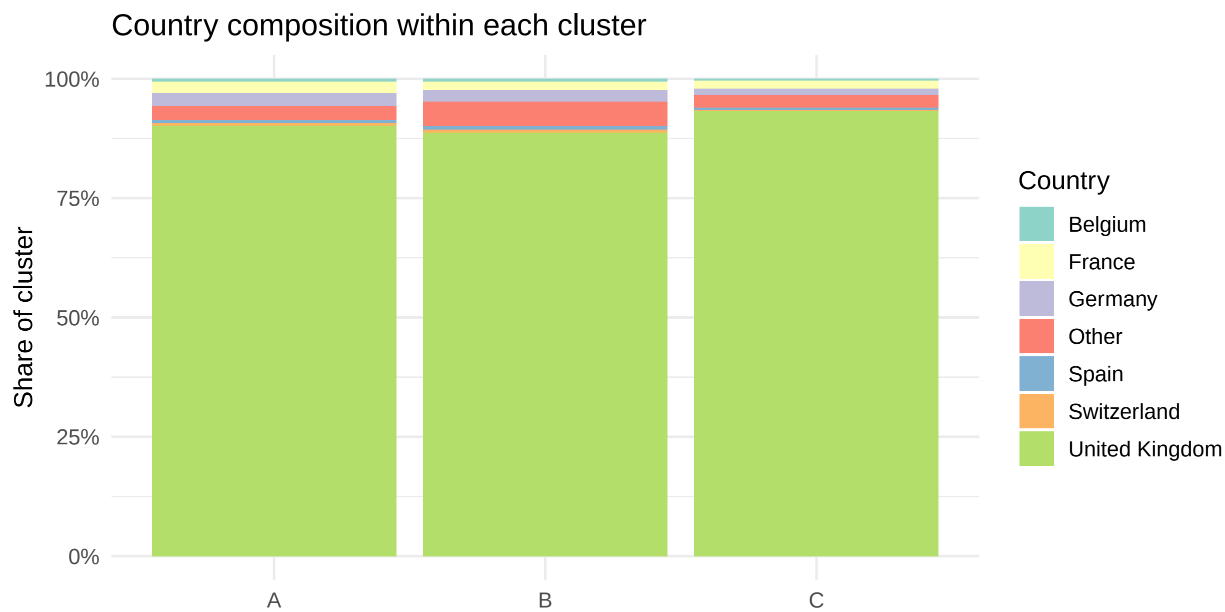
```r
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_colour_brewer(palette = "Set1") +
  labs(title = "Cluster signature   feature means in standard deviations",
       x = NULL, y = "z-score") +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



7. Geographic composition

`Country` was held out of clustering on purpose (UK ~ 83%). Using it now is descriptive only it tells us whether segments correlate with geography after the fact.

```r
top_countries <- cust |> dplyr::count(Country, sort = TRUE) |>
  slice_max(n, n = 6) |> pull(Country)

cust |>
  mutate(Country = ifelse(Country %in% top_countries, Country, "Other")) |>
  dplyr::count(cluster, Country) |>
  group_by(cluster) |>
  mutate(share = n / sum(n)) |>
  ggplot(aes(cluster, share, fill = Country)) +
  geom_col() +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_brewer(palette = "Set3") +
  labs(title = "Country composition within each cluster",
       y = "Share of cluster", x = NULL)
```



8. Top products per cluster with AND without anomaly

```
anomaly_code <- "23843"

top_with <- tx |>
  group_by(cluster, StockCode, Description) |>
  summarise(revenue = sum(Amount), .groups = "drop") |>
  group_by(cluster) |>
  slice_max(revenue, n = 5) |>
  ungroup()

top_with |>
  mutate(Description = str_trunc(Description, 35),
         revenue     = dollar(revenue, prefix = pound)) |>
  kable(caption = "Top 5 products by revenue, per cluster (INCLUDING 23843)")
```

Table 3: Top 5 products by revenue, per cluster (INCLUDING 23843)

cluster	StockCode	Description	revenue
A	22423	REGENCY CAKESTAND 3 TIER	£120,435
A	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£75,188
A	85099B	JUMBO BAG RED RETROSPOT	£73,401
A	47566	PARTY BUNTING	£52,392
A	84879	ASSORTED COLOUR BIRD ORNAMENT	£45,368
B	23843	PAPER CRAFT , LITTLE BIRDIE	£168,470
B	23166	MEDIUM CERAMIC TOP STORAGE JAR	£77,942
B	22502	PICNIC BASKET WICKER 60 PIECES	£39,620
B	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£20,844

cluster	StockCode	Description	revenue
B	22423	REGENCY CAKESTAND 3 TIER	£15,386
C	22423	REGENCY CAKESTAND 3 TIER	£6,444
C	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£4,361
C	47566	PARTY BUNTING	£2,928
C	85099B	JUMBO BAG RED RETROSPOT	£2,774
C	48138	DOORMAT UNION FLAG	£2,699

```r
top_without <- tx |>
  filter(StockCode != anomaly_code) |>
  group_by(cluster, StockCode, Description) |>
  summarise(revenue = sum(Amount), .groups = "drop") |>
  group_by(cluster) |>
  slice_max(revenue, n = 5) |>
  ungroup()

top_without |>
  mutate(Description = str_trunc(Description, 35),
         revenue     = dollar(revenue, prefix = pound)) |>
  kable(caption = "Top 5 products by revenue, per cluster (EXCLUDING 23843)")
```

Table 4: Top 5 products by revenue, per cluster (EXCLUDING 23843)

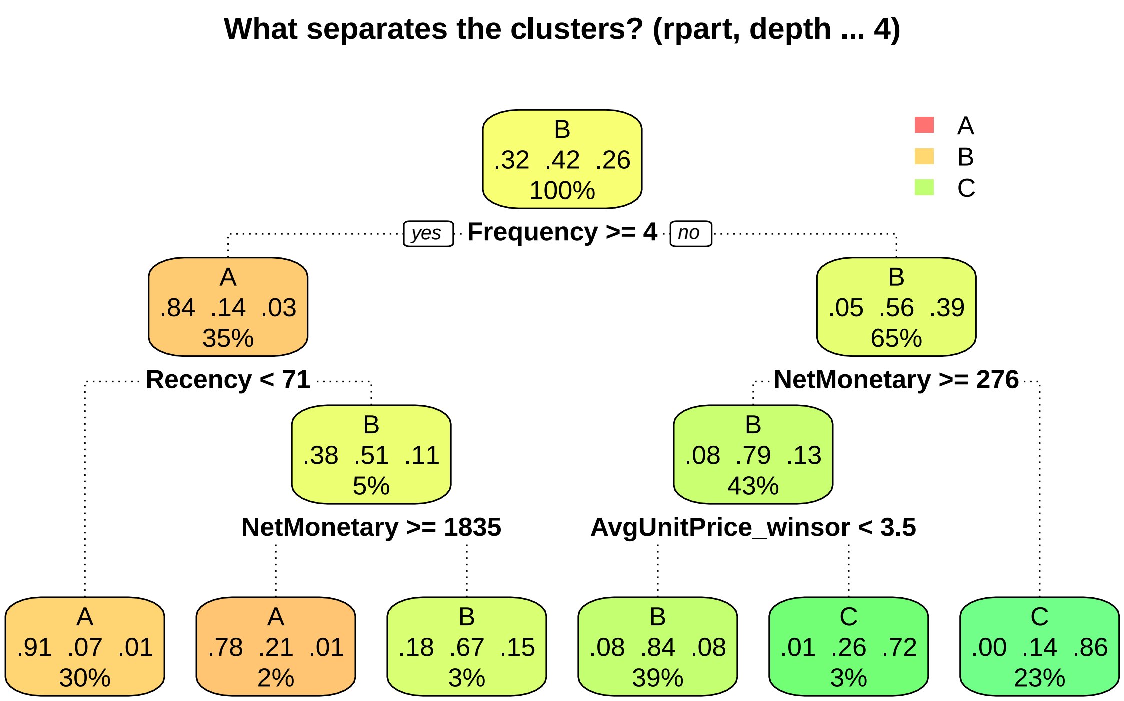
cluster	StockCode	Description	revenue
A	22423	REGENCY CAKESTAND 3 TIER	£120,435
A	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£75,188
A	85099B	JUMBO BAG RED RETROSPOT	£73,401
A	47566	PARTY BUNTING	£52,392
A	84879	ASSORTED COLOUR BIRD ORNAMENT	£45,368
B	23166	MEDIUM CERAMIC TOP STORAGE JAR	£77,942
B	22502	PICNIC BASKET WICKER 60 PIECES	£39,620
B	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£20,844
B	22423	REGENCY CAKESTAND 3 TIER	£15,386
B	21108	FAIRY CAKE FLANNEL ASSORTED COLOUR	£14,622
C	22423	REGENCY CAKESTAND 3 TIER	£6,444
C	85123A	WHITE HANGING HEART T-LIGHT HOLDER	£4,361
C	47566	PARTY BUNTING	£2,928
C	85099B	JUMBO BAG RED RETROSPOT	£2,774
C	48138	DOORMAT UNION FLAG	£2,699

The two tables together tell the anomaly story directly: if `23843` appears as a single cluster's top product in the first table but disappears entirely from the second, that cluster's "top product" was really a single bulk wholesale invoice (see §9 of `01_EDA.Rmd`). The second table is the more honest description of the segment's habitual product mix.

9. What separates the clusters? a 1-page decision tree

Centroids and parallel coordinates show *where* clusters sit. A small classification tree shows *which features and which thresholds* you would actually use to assign a new customer to a cluster. We fit `rpart` on the original (un-PCA'd) features so the splits are interpretable in business units (£, days, counts) rather than PC scores.

```
tree_data <- cust |>
  select(cluster,
         Recency, Frequency, NetMonetary,
         AvgOrderValue, UniqueProducts,
         AvgUnitPrice_winsor, TenureDays)

tree <- rpart(cluster ~ ., data = tree_data,
              method = "class",
              control = rpart.control(cp = 0.01, maxdepth = 4,
                                      minbucket = 50))

rpart.plot(tree,
           type = 2, extra = 104, fallen.leaves = TRUE,
           box.palette = "RdYlGn", branch.lty = 3,
           main = "What separates the clusters? (rpart, depth   4)")
```




```
pred <- predict(tree, tree_data, type = "class")
acc  <- mean(pred == tree_data$cluster)

list(
  in_sample_accuracy = round(acc, 3),
  confusion          = table(predicted = pred, actual = tree_data$cluster)
)
```

```
## $in_sample_accuracy
```

```
## [1] 0.855
##
## $confusion
##          actual
## predicted    A    B    C
##         A 1227  108   17
##         B  167 1522  159
##         C    2  174  958
```

The tree is descriptive, not predictive: the labels themselves came from k-means, so the in-sample accuracy is an upper bound on how recoverable the clustering is from the original features. A tree accuracy above ~0.85 means the clusters can be substantially explained by 3–4 simple rules on the raw features, which is the precondition for naming them in §10.

10. Persona names

The persona names below are the human-readable labels that go on dashboards, briefs, and CRM segments. They are derived directly from the §4 profile table and the §9 tree splits no inputs that aren't already in this notebook.

Cluster	Persona	One-line description
A	**Wholesale-leaning regulars**	High net spend, frequent orders, broadest product range, often higher AvgOrderValue. Small share of customers, large share of revenue.
B	**Engaged retail**	Recent and reasonably frequent buyers at typical retail order sizes. Largest share of customers, moderate share of revenue.
C	**Lapsed / one-time**	High recency (haven't bought in a long time), low frequency, often `SinglePurchase = TRUE`. Large share of customers, small share of revenue.

These names are consistent across the assignment-stable sense of §6 of `03_clustering.Rmd`: persona **A** corresponds to the high-Jaccard high-value cluster and is safe to act on; persona **B** sits in the fuzzier middle band and should be targeted with broader campaigns; persona **C** is presented as "directional only" wherever the bootstrap Jaccard came in below 0.75, because the boundary between **B** and **C** is the part of the partition that moves most under resampling.

```
persona_lookup <- tibble(
  cluster = factor(c("A", "B", "C"),
                   levels = LETTERS[seq_along(levels(cust$cluster))]),
  persona = c("Wholesale-leaning regulars",
              "Engaged retail",
              "Lapsed / one-time")
) |>
  filter(cluster %in% levels(cust$cluster))
```

```r
cust <- cust |>
  left_join(persona_lookup, by = "cluster")

table(cust$cluster, cust$persona)
```

```
##
##     Engaged retail Lapsed / one-time Wholesale-leaning regulars
##   A              0                 0                       1396
##   B           1804                 0                          0
##   C              0              1134                          0
```

11. Executive summary

Three numbers the executive summary should lead with:

```r
exec <- cust |>
  group_by(cluster, persona) |>
  summarise(
    customers     = n(),
    pct_customers = n() / nrow(cust),
    revenue_net   = sum(NetMonetary),
    pct_revenue   = sum(NetMonetary) / sum(cust$NetMonetary),
    median_recency_d = median(Recency),
    .groups = "drop"
  ) |>
  arrange(cluster)

exec |>
  mutate(revenue_net   = dollar(revenue_net,  prefix = pound, accuracy = 1),
         pct_customers = percent(pct_customers, accuracy = 0.1),
         pct_revenue   = percent(pct_revenue,   accuracy = 0.1)) |>
  kable(caption = "Executive summary   customers, revenue, recency by persona")
```

Table 6: Executive summary customers, revenue, recency by persona

cluster	persona	customers	pct_customers	revenue_net	pct_revenue	median_recency_d
A	Wholesale-leaning regulars	1396	32.2%	£6,574,443	79.5%	16.0
B	Engaged retail	1804	41.6%	£1,447,805	17.5%	70.5
C	Lapsed / one-time	1134	26.2%	£247,152	3.0%	156.0

The recommended actions follow directly from the customer-share / revenue-share asymmetry:

- **Persona A protect.** Disproportionate revenue from a small customer base. Lifecycle programmes, account-management touchpoints, early access to new SKUs. Loss aversion: the cost of churning one A is the cost of acquiring many Bs.
- **Persona B convert.** Largest customer base, moderate revenue per head. Treatment effect leverage is highest here an offer that shifts a B toward A behaviour is worth more in expectation than the same offer to an A. Test promotions, cross-sell adjacencies, and basket-size nudges.

- **Persona C selective re-engagement.** Don't blanket-target a segment with low Jaccard stability. Run a controlled win-back to the most recently lapsed slice of C and measure incrementality before scaling.